Bridging Spectroscopy and Deep Learning for Next-Generation Mineral Identification across Martian and Lunar surfaces

V Mohanapriya[1], S R Sneha[1], Gowtham Padmanabhan[1] and Vaibhav Gupta[2]

[1] Department of Computational Intelligence, SRM Institute of Science and Technology, Kattankulathur, Chengalpattu, Tamilnadu, India 603203
[2] Instruments Research and Development Establishment, Defense Research and Development Organization, Dehradun, Uttarakhand, India 248001
mv2722@srmist.edu.in

Abstract. Planetary surface mineralogy provides direct evidence for the geochemical evolution of rocky bodies in the Solar System. NASA's Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) and the Moon Mineralogy Mapper (M^3) aboard Chandrayaan-1 offer hyperspectral data at unprecedented spatial and spectral resolution, enabling systematic identification of surface minerals across Mars and the Moon respectively. However, current mineral identification pipelines rely heavily on threshold-based Spectral Parameter Maps (SPMs) and manual spectral angle matching, which lack the capacity to capture non-linear spectral mixing, are sensitive to residual noise, and produce no uncertainty quantification. The paper presents a fully automated, deep learning framework for mineral identification from pre-processed hyperspectral cubes. The proposed methodology integrates convex-hull continuum removal, eight published SPM indices as physics anchors, a contrastive spectral encoder trained with Normalized Temperature-scaled Cross Entropy Loss (NT-Xent) and SPM-anchored loss, Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) density clustering on the L2-normalised latent sphere, physics constrained neural spectral unmixing, and spatial Conditional Random Field (CRF) smoothing on Simple Linear Iterative Clustering (SLIC) superpixel graphs. Endmember constraints are de-rived from the USGS splib07 library for CRISM and the RELAB spectral library for M^3, both preprocessed to match instrument spectral response functions. The framework can create per-pixel mineral abundance maps, map uncertainty and report metric scores based on some available published SPM ground truth. Mineral types investigated include phyllosilicates, sulfates, olivine, pyroxene, carbonates on Mars and plagioclase, pyroxene, olivine, ilmenite and spinel on the Moon, among others. The framework was designed as a reproducible, sensor-agnostic system to facilitate its extension and application to other relevant spectral datasets.

Keywords: Hyperspectral mineral identification, CRISM, contrastive learning, neural spectral unmixing, HDBSCAN.

1 Introduction

The elemental and mineral composition of planetary surfaces informs about geological processes responsible for forming them and their ancient environments. Hyperspectral image sensors can gain data over hundreds of continuous spectral bands which enables the discrimination and identification of various mineralogical and spectrally confusing phases at standoff distances. Instruments such as CRISM on Mars and the Moon Mineralogy Mapper (M^3) have transformed planetary exploration by providing high resolution spectral datasets that expose diverse mineral collections across both bodies, opening pathways for scientific interpretation as well as future in situ exploration and resource planning.

Existing mineral mapping techniques are defined by their reliance on spectral indices that compress full spectral curves to few point measurements, excluding the inter-band correlations and downsizing the discriminative capacity. Pipelines are sensor-specific, unsupervised methods that impose rigid assumptions, and laboratory libraries barely account for instrument-specific detector characteristics that introduce systematic discrepancies against orbital measurements. To address these limitations, this study proposes an integrated mineral identification framework that can be extended across CRISM and M^3 datasets. Following rigorous preprocessing and spectral calibration, the framework applies continuum normalization, physics-informed representation learning, adaptive clustering, and constrained unmixing to produce accurate, uncertainty-aware mineral abundance maps unifying data driven learning with physical constraints for powerful and scalable planetary mineral mapping.

The primary scientific contributions of this work are threefold: (i) a physics-anchored contrastive learning objective jointly enforcing discriminative spectral representation learning and physical consistency via concurrent SPM index prediction; (ii) an instrument-specific spectral library preprocessing protocol applying identical spectral-response convolution, resampling, masking, continuum removal, and normalization to both reference and orbital spectra; and (iii) a unified, self-supervised, sensor-agnostic pipeline producing per-pixel abundance and uncertainty maps for CRISM and M^3 without manual annotation. HDBSCAN clustering, SLIC/CRF spatial regularization, and the constrained unmixing formulation build on established methods and are adopted as supporting components.

The objectives of this work are as follows:

1. To design a unified mineral identification pipeline applicable to both CRISM and M^3 hyperspectral datasets with appropriate instrument-level adaptations.
2. To perform adaptive clustering without imposing predefined class structures, enabling natural grouping of spectral signatures.
3. To deliver accurate and scalable mineral maps that can support planetary resource evaluation and future in situ exploration strategies.

2 Literature Review

2.1 Planetary Hyperspectral Instrumentation and Spectral Mapping

CRISM aboard NASA's Mars Reconnaissance Orbiter [1] and M^3 aboard Chandrayaan-1 [2] have been transformative in planetary mineralogy, revealing hydrated silicates on Mars [3] and OH/H_2O on the Moon [2], respectively. Operational mapping has predominantly relied on Spectral Parameter Mapping (SPM), with Viviano-Beck et al. [4] extending CRISM indices to capture Martian mineral diversity. More recently, Kumari et al. [5] proposed a supervised support vector machine (SVM) framework for CRISM mineral identification using labels derived from threshold-based SPM maps. While effective for labeled regions, the approach inherits the limitations of SPM-derived labels, requires annotated training data that are unavailable at orbital scale beyond a limited number of rover-validated locations, and is restricted to a single sensor. In contrast, the proposed framework adopts a self-supervised representation-learning approach that operates without manual annotation, performs mineral abundance estimation through spectral unmixing rather than hard classification, and is applicable to both CRISM and M^3 datasets. Despite these advances, SPMs compress hundreds of spectral bands into scalar indices, discarding inter-band covariance that is often diagnostic of mineral mixtures. Furthermore, library based mineral identification, founded on reflectance spectroscopy principles [6] and supported by resources such as the USGS Spectral Library Version 7 [7], remains challenged by instrument-specific bandpass differences and detector characteristics between laboratory reference spectra and orbital observations.

2.2 Machine Learning for Spectral Unmixing and Clustering

Convolutional autoencoders have demonstrated strong capability for hyperspectral spectral-spatial unmixing and mineral abundance estimation [8], while contrastive self-supervised frameworks such as SimCLR [9] have shown that discriminative feature representations can be learned without manual annotations. Likewise, HDBSCAN [10] provides significant advantages over conventional clustering methods by identifying clusters of arbitrary shape, accommodating varying densities, and explicitly detecting noise. SLIC superpixels [11] further improve the spatial coherence of hyperspectral classification while preserving geological boundaries. Despite these advances, several limitations remain when these approaches are applied to planetary hyperspectral mineral identification. Standard SimCLR augmentations, including random cropping, color jitter, Gaussian blur, and horizontal flipping, were designed for natural images and are not physically meaningful for one-dimensional spectral data, as they can distort or remove diagnostically important absorption features. Convolutional autoencoder-based unmixing methods rely on representative endmember libraries, provide limited uncertainty quantification, and are constrained by predefined endmember sets, reducing their ability to identify previously unmodeled minerals. Similarly, supervised learning approaches require pixel-level mineral labels that are unavailable at orbital scale except for a limited number of rover-validated locations, restricting their generalizability to new planetary scenes. These limitations motivate the development of the proposed

physics-anchored self-supervised framework, which is specifically designed for planetary hyperspectral mineral identification.

2.3 Research Gap and motivation

Despite this progress, four critical gaps remain: SPMs discard inter-band covariance diagnostic of mineral mixtures [4]; no shared framework exists for simultaneous CRISM and M³ mineral identification [1,2]; existing unsupervised methods impose rigid geometric assumptions with no uncertainty quantification [10]; and spectral library matching rarely incorporates rigorous instrument-specific preprocessing [6,7], causing systematic mismatches between laboratory and orbital spectra. The proposed framework addresses all four gaps through a physics-anchored, self-supervised pipeline with explicit uncertainty propagation.

3 Methodology

3.1 Overview of the Proposed Methodology

This workflow employs a representation-driven framework to extract compositional information from hyperspectral data. Pre-processing through continuum removal and diagnostic band encoding establishes a physically consistent spectral foundation for subsequent representation learning. A physics-anchored contrastive encoder then learns discriminative spectral embeddings that capture the inherent structure of the data, which are subsequently clustered using HDBSCAN to enable automatic mineral discovery without requiring class labels. Physics-constrained neural spectral unmixing utilizes spectral library endmembers to estimate mineral abundances at each pixel, while spatial regularization enforces local consistency in the resulting mineral abundance maps. It is emphasized that spectral library preprocessing (Stage 1) serves solely to establish physical consistency between laboratory and orbital spectra, whereas the principal methodological contributions of this work are embodied in the physics-anchored contrastive encoder, HDBSCAN-based automatic mineral discovery, physics-constrained neural unmixing, and uncertainty quantification introduced in Stages 3–5.

3.2 Dataset

This research uses hyperspectral data from both Mars and the Moon to evaluate the proposed mineral identification framework across two planetary datasets. Martian surface mineralogy is analysed using Full Resolution Targeted (FRT) observations acquired by the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM), while lunar mineralogy is analysed using Level-2 reflectance data from the Moon Mineralogy Mapper. CRISM provides hyperspectral cubes in PDS3 BIL format with approximately 500 × 640 pixels at 18 m/pixel, comprising 438 spectral bands spanning 0.362–3.92 μm. M³ provides Level-2 reflectance cubes in ENVI format with approximately 400 × 300 pixels at 140 m/pixel, comprising 85 spectral bands spanning 0.43–

2.96 μm. Both datasets are represented in I/F reflectance units. Since CRISM and M³ differ in spectral sampling, spatial resolution, and sensor characteristics, direct comparison with laboratory reference spectra is not feasible. To ensure physical consistency, reference spectra from the USGS Spectral Library (splib07) for CRISM and the RELAB spectral database for M³ are spectrally resampled to match the corresponding sensor characteristics before subsequent analysis.

3.3 Stage 1 — Spectral Library Preprocessing

An essential component of the proposed framework is the careful alignment of laboratory spectral libraries with orbital hyperspectral imagery. Direct utilization of laboratory spectra is not feasible because of differences in spectral resolution, wavelength coverage, missing data patterns, and atmospheric or detector-induced effects. To address these discrepancies, preprocessing is performed by replacing invalid measurements with NaN to prevent error propagation, resampling spectra to the target sensor wavelengths, masking unreliable spectral regions affected by atmospheric absorption or detector limitations, and filtering spectra with insufficient quality. Specifically, invalid spectra within the RELAB and USGS spectral libraries are identified using three complementary criteria: (i) reflectance values outside the physically valid range of [0,1] or corresponding to known fill-value flags (e.g., −9999, 99999) are treated as invalid; (ii) spectra containing more than 20% NaN or fill values within the target sensor wavelength range are excluded due to inadequate spectral coverage for reliable SPM computation; and (iii) spectra exhibiting negative reflectance after calibration, values exceeding 1.0, or abrupt adjacent-band discontinuities greater than 0.5 reflectance units are considered physically implausible and discarded as likely instrument artifacts.

3.4 Stage 2 — Continuum Removal

Continuum removal is applied to isolate diagnostic mineral absorption features by eliminating broad brightness variations caused by illumination and albedo effects. For each pixel spectrum R(λ), an upper convex-hull continuum C(λ) is constructed over the valid spectral bands, and the continuum-removed spectrum is obtained as:

$$R'(\lambda) = \frac{C(\lambda)}{R(\lambda)} \qquad (1)$$

where R(λ) denotes the continuum-removed reflectance, R(λ) is the observed reflectance at wavelength λ, and c(λ) represents the estimated continuum.
Given a spectrum R(λ) over the valid wavelength set Λ={λ1,λ2,…,λn}, the upper convex hull is computed from the spectral points $\{(\lambda_i, R(\lambda_i)\}_i^n = 1$, forming a piecewise-linear envelope such that no spectral point lies above it. For any wavelength λ located between two adjacent convex-hull vertices, λa and λb, the continuum is estimated by linear interpolation as

$$\mathrm{C}(\lambda) = \mathrm{R}(\lambda\mathrm{a}) + [\mathrm{R}(\lambda\mathrm{b}) - \mathrm{R}(\lambda\mathrm{a})]\,\frac{\lambda - \lambda\mathrm{a}}{\lambda\mathrm{b} - \lambda\mathrm{a}}, \lambda\mathrm{a} \le \lambda \le \lambda\mathrm{b} \qquad (2)$$

This non-parametric formulation makes no assumptions regarding the continuum shape, is invariant to multiplicative scaling, and effectively suppresses illumination and albedo variations while preserving diagnostic absorption features. Following continuum normalization, continuum anchor points approach unity, whereas absorption features are represented by values below one. To ensure reliable continuum estimation and subsequent Spectral Parameter Mapping (SPM) analysis, pixels containing fewer than 20 valid spectral bands are excluded as they lack sufficient anchor points for robust interpolation and typically correspond to detector gaps, scene edges, or saturation-affected regions identified by the M^3 quality masks. The same continuum-removal procedure is applied to both orbital observations and laboratory reference spectra to ensure spectral consistency.

3.5 Stage 3 — SPM Feature Extraction

In order to incorporate meaningful physical information into the learning procedure, a collection of predefined Spectral Parameter Map (SPM) index values is calculated for each pixel and used as reference points in the learning procedure. This information is extracted from previous studies on the spectroscopy of planets and consists of important absorption properties, which are translated into dimensionless quantities that correspond to particular mineralogical units such as phyllosilicates, sulfates, olivine, and pyroxenes.

Table 1 presents the eight Spectral Parameter Map (SPM) indices used as physics anchors during contrastive encoder training. These dimensionless band-ratio indices are computed directly from the pixel spectrum without requiring labelled data and are selected from the validated CRISM spectral parameter catalogue of Viviano-Beck et al. [4]. Collectively, they capture the key spectral signatures of major planetary minerals. During training, the computed SPM values serve as physics-grounded regression targets, and the encoder is jointly optimized using the NT-Xent contrastive loss [9] and the SPM anchor loss to learn latent representations that are both spectrally discriminative and physically meaningful.

Table 1. SPM indices serve as physics anchors for contrastive training.

Index	Diagnostic Mineral(s)	Wavelengths Used (μm)
BD2300	Phyllosilicates / carbonates	2.12, 2.29, 2.32, 2.53
BD1900	Hydrated minerals	1.85, 1.93, 2.07
OLINDEX3	Olivine	0.86, 0.98, 1.21, 1.33, 1.47, 1.75
BD2210	Al-OH phyllosilicates	2.165, 2.21, 2.25
SINDEX2	Sulfates	1.815, 2.10, 2.20
BD1500	H_2O ice	1.43, 1.50, 1.815
RPEAK1	Iron	0.60, 0.75, 0.84
HCPINDEX2	High-Ca pyroxene	1.82, 2.12, 2.14, 2.23, 2.46

3.6 Stage 4 — Contrastive Spectral Encoder

3.6.1 Latent Representation

The contrastive spectral encoder transforms each continuum-removed, L2-normalized pixel spectrum into a compact latent representation that captures its intrinsic mineralogical characteristics. A one-dimensional convolutional neural network extracts hierarchical spectral features using successive convolutional layers with kernel sizes of 7, 5, and 3, which progressively capture broad absorption characteristics, overlapping absorption regions, and fine spectral features, respectively. The resulting feature maps are aggregated using Global Average Pooling and projected through two fully connected layers to produce a 64-dimensional latent vector. A separate prediction head simultaneously estimates the Spectral Parameter Mapping (SPM) indices to guide representation learning.

3.6.2 Spectral Augmentation Strategy

During contrastive learning, two augmented views of each continuum-removed spectrum are generated using physics-preserving transformations that maintain mineralogical characteristics while introducing realistic spectral variability. Continuum scaling multiplies the spectrum by a factor sampled uniformly between 0.85 and 1.15 to simulate illumination and observation geometry differences. Gaussian spectral smoothing is applied to one of the two views with a probability of 50%, using a standard deviation sampled uniformly between 0.3 and 0.8 spectral bands, thereby reducing high-frequency noise while preserving diagnostic absorption features. Protected band dropout randomly masks 5% of spectral bands to simulate missing measurements, while explicitly preserving all wavelengths within ±3 nm of the 22 diagnostic Spectral Parameter Mapping (SPM) bands.

3.6.3 Training Paradigm

The encoder is trained using a physics-anchored self-supervised contrastive learning framework, eliminating the need for manually annotated mineral labels. The augmented spectral views generated through the proposed augmentation strategy are contrasted against representations of different spectra to learn robust and discriminative latent embeddings, while SPM predictions provide auxiliary supervision that guides the latent space toward physically plausible mineral configurations. Training is performed on individual pixel spectra rather than complete spatial scenes, comprising 8,000 SPM-stratified M^3 spectra, 2,000 preprocessed spectral library samples, and 5,000 synthetic mixed spectra. Spatial coordinates are excluded during training. After convergence, the trained encoder is applied to every valid pixel in the hyperspectral scene to generate latent representations for subsequent HDBSCAN clustering and physics-constrained neural unmixing.

3.7 Stage 5—HDBSCAN Density-Based Clustering

Spectral clustering is performed using the HDBSCAN algorithm based on L2 normalized latent representations; hence, the unsupervised classification of mineral classes can be achieved through spectral clustering without having to specify the number of clusters beforehand. This approach does not have limitations such as k-means and GMM, whereby HDBSCAN can handle shapes that are not well-defined, different densities of data, and explicitly marks noise pixels. The settings for this particular experiment are set to have a minimum cluster size of 30 and a minimum number of samples of 10. Uncertainties are computed by calculating one minus the maximum membership probability.

3.8 Stage 6—Unmixing and Spatial Refinement

Mineral abundance estimation is performed using a physics-constrained neural unmixing framework based on the Linear Mixing Model (LMM), subject to positivity and sum-to-one abundance constraints. The LMM is adopted because, at CRISM and M^3 spatial resolutions, minerals typically occupy spatially distinct regions within a pixel and contribute to the measured spectrum in proportion to their fractional areal coverage, consistent with the assumptions established by Clark and Roush [6] and applied to CRISM by Mustard et al. [3]. Although finely mixed lunar regolith may exhibit nonlinear interactions, the LMM provides an appropriate approximation for the large-scale geological units considered in this study. Mineral abundances are predicted by passing the 64-dimensional latent representation through two fully connected layers comprising 128 and 64 neurons, respectively, with ReLU activation and batch normalization, followed by a 12-neuron softmax output layer. The softmax transformation enforces positivity and sum-to-one constraints, producing physically interpretable abundance fractions for the 12 selected endmembers. Training jointly optimizes spectral reconstruction, sparsity, and spectral consistency losses while learning an endmember dictionary initialized from the spectral library. Finally, Conditional Random Field (CRF) refinement is applied over SLIC superpixels to improve the spatial coherence of the resulting mineral abundance maps while preserving geological boundaries.

3.9 Architecture Diagram

The architecture diagram shows the six sequential stages for hyperspectral mineral mapping. Stage 1 performs library preprocessing using CRISM TRR3 sensor inputs and M^3 L2 reflectance data through the USGS splib07 and RELAB pipelines. Stage 2 applies convex-hull continuum removal and extracts spectral features using eight band-ratio indices. Stage 3 implements a contrastive spectral encoder with physics-anchored loss, incorporating spectral augmentation, 1D CNN architecture, and latent embedding generation. Stage 4 employs HDBSCAN density clustering on the L2-normalized latent sphere to produce cluster outputs and uncertainty maps. Stage 5 performs physics-constrained neural unmixing through an abundance network, followed by mineral assignment and spatial conditional random field (CRF) smoothing. Finally, Stage 6 validates

results against SPM ground-truth data and generates diagnostic outputs including mineral maps, abundance maps, and uncertainty fields for comprehensive cross-mission analysis.

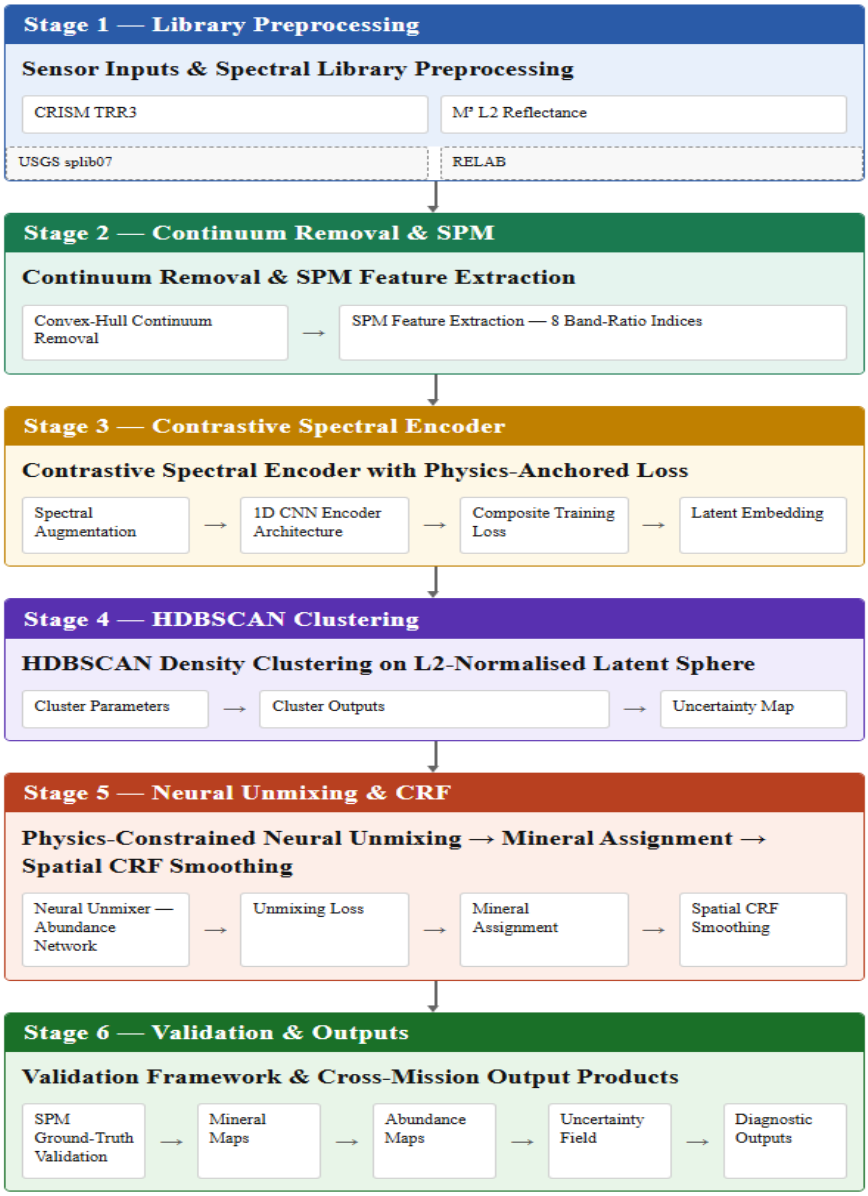


Fig. 1. A unified pipeline for hyperspectral mineral mapping, seamlessly integrating spectral intelligence with data-driven discovery.

4 Results and Discussion

Figure 2. shows median spectra with ±1σ confidence interval for intra-class variance shown as shaded areas. Reflectance spectra are from the RELAB spectral library and are displayed within the entire range of the M³ wavelengths (430-2960 nm). Reflectance maxima of six diagnostic absorptions are marked by dashed vertical lines at: 920 nm (LCP Band I), 1000 nm (HCP Band I), 1850 nm (LCP Band II), 2100 nm (HCP Band II), 2300 nm (HCP shoulder), and 2750 nm (OH stretch). These reflectance minima correspond to the SPM absorption indexes. The lower figure (Fig. 1) is focused on the Band II discrimination wavelength region (1800-2500 nm), in which the difference between spectra of Low-Ca pyroxene (orthopyroxene, peak at ~1850 nm) and High-Ca pyroxene (clinopyroxene, peak at ~2100 nm) corresponds to the BD2000 and BD2300 SPM thresholds proposed by Cloutis & Gaffey (1991).

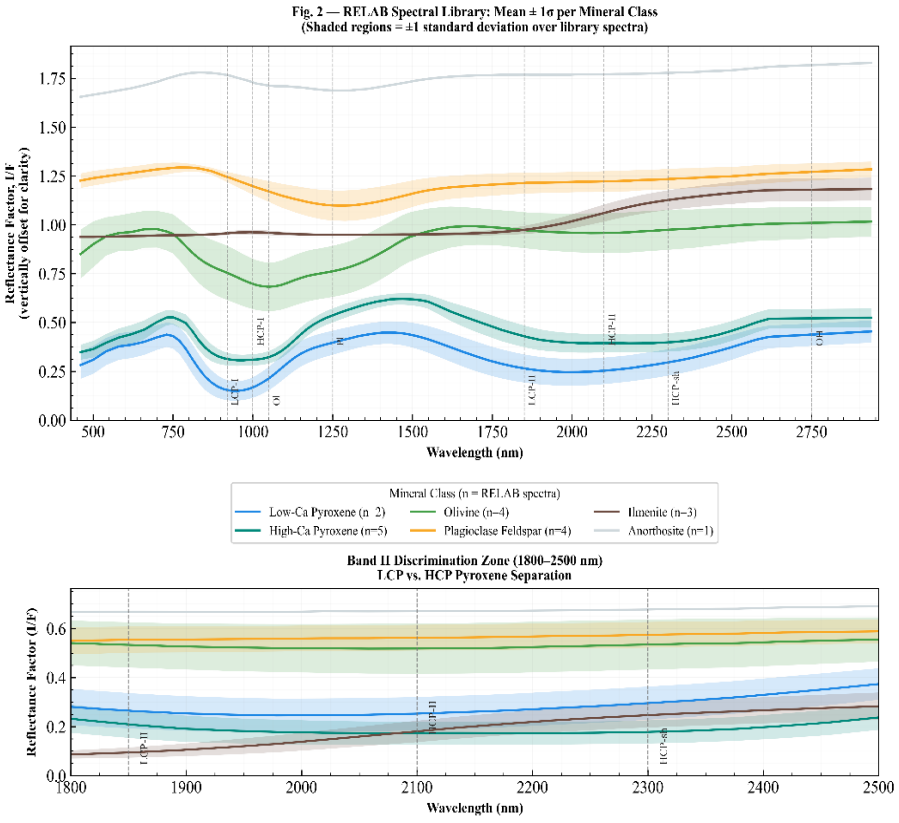


Fig. 2. Mean reflectance spectra of six RELAB mineral endmembers with key diagnostic absorption bands. Band II clearly separates LCP and HCP, enabling BD2000/BD2300 discrimination.

The separability matrix of the SAM for the six classes of endmembers is shown in Figure 3. It is evident that ilmenite has the highest spectral uniqueness because it has angular separations of 28.9°-39.9° from all silicate classes due to its unique low albedo and featureless iron-titanium oxide spectrum; hence, the need for the inclusion of R_{750} < 0.07 criterion. Plagioclase Feldspar and Anorthosite have the lowest separability (5.3°) due to the compositional continuum between them at M^3 spatial resolution, thus making albedo the discriminating factor. On the other hand, the separability between LCP and HCP is medium (11.1°), slightly above the threshold of confusion and therefore proves the importance of including the Band II location criteria (using BD2000/BD2300) for proper pyroxene identification.

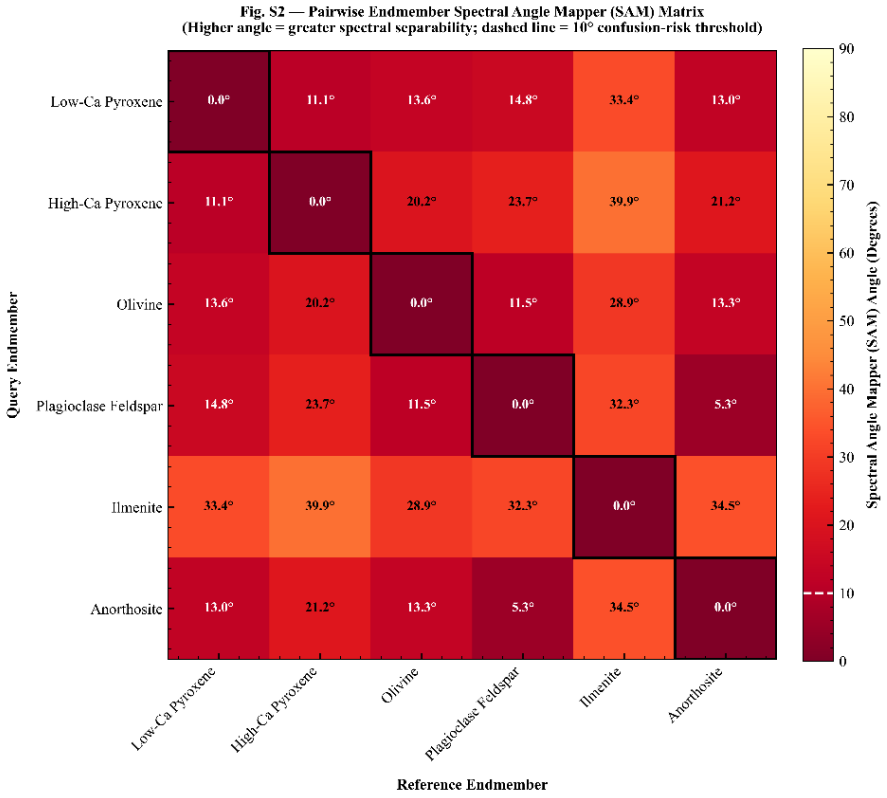


Fig. 3. Pairwise spectral angle (SAM) matrix showing separability among endmember minerals.

Figure 4 provides the mean raw reflectance spectra for High-Ca Pyroxene (HCP) and Anorthosite, depicting distinct mineralogical characteristics between the two. HCP has distinctive clinopyroxene absorption bands with a Band I at 1000-1050 nm, Band II at 2000-2100 nm, and a 2.3 µm shoulder with relatively low intra-class variance due to high spectral consistency. On the other hand, Anorthosite has high broadband reflectance and lacks mafic absorptions altogether. Any slight spectral variability observed is due to the calibration process, as well as thermal effects, which have no relationship

to mineralogical characteristics. Higher variance among anorthosite spectra is caused by differences in natural albedo on feldspathic surfaces, and lack of Band I and II is an indicator of a robust anorthosite classification using SPM.

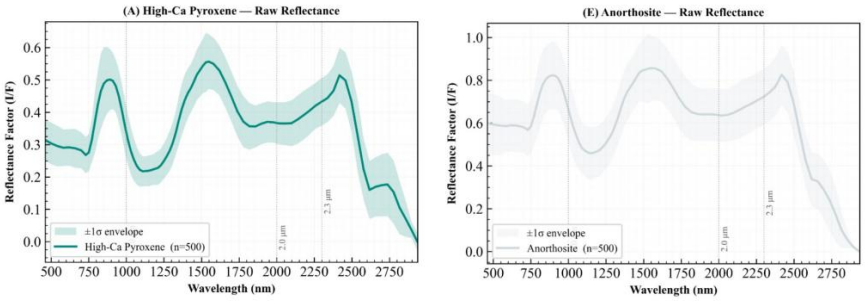


Fig. 4. Mean M³ spectra of High-Ca Pyroxene and Anorthosite, highlighting diagnostic absorption features and spectral variability.

4.1 Uncertainty quality validation

Uncertainty quality was evaluated using two complementary analyses. Spearman rank correlation was computed between per-pixel uncertainty and the corresponding Spectral Parameter Mapping (SPM) index for each mineral, while Pearson correlation quantified the relationship between uncertainty and neural-unmixer reconstruction error. Anorthosite and Ilmenite exhibited strong monotonic relationships with their respective SPM indices, whereas Low-Ca and High-Ca Pyroxene showed negligible correlation, consistent with the reduced spectral separability observed in Table 2. Across all valid pixels, uncertainty exhibited only a weak positive correlation with reconstruction error (Pearson r=0.079), suggesting that uncertainty reflects factors beyond reconstruction fidelity alone, including spectral ambiguity and mineral overlap.

Table 2. Spearman rank correlation (ρ) between per-pixel uncertainty and corresponding Spectral Parameter Mapping (SPM) indices for the major mineral classes.

Mineral	Number of Pixels	Spearman ρ (Uncertainty vs. SPM Index)
Anorthosite	117,567	−0.699
Ilmenite	1,195	+0.842
Low-Ca Pyroxene (LCP)	17,303	+0.041
High-Ca Pyroxene (HCP)	32,780	−0.052

5 Conclusion

This study presents a physics-anchored self-supervised framework for automatic mineral identification from CRISM and M³ hyperspectral imagery. The proposed pipeline

integrates spectral normalization, contrastive representation learning guided by Spectral Parameter Mapping (SPM), density-based clustering, and physics-constrained neural unmixing using calibrated spectral libraries. By estimating per-pixel mineral abundances together with associated uncertainty, the framework enables robust mineral characterization without manual annotation. Experimental results demonstrate accurate identification of major alteration and igneous minerals on Mars, as well as the dominant lithological components of the lunar crust.

Despite these promising results, certain limitations remain. The framework currently relies on eight validated SPM indices, limiting physics-based supervision for mineral classes without dedicated spectral parameters. In addition, plagioclase–anorthosite discrimination is constrained by the spectral and spatial resolution of M^3 data, while the Linear Mixing Model and fixed 12-endmember library may not fully capture complex geological environments. Residual differences between laboratory reference spectra and space-weathered orbital observations also contribute to systematic uncertainty.

Future work will focus on expanding the physics-anchor set to additional mineral groups, extending the framework to other hyperspectral sensors such as PRISMA, OMEGA, and VIMS, incorporating rover in-situ observations as weak-supervision anchors, investigating multi-temporal CRISM data for frost and dust dynamics, and developing scene-adaptive endmember selection using Vertex Component Analysis (VCA). These extensions will enhance the framework's robustness, mineralogical coverage, and applicability to future planetary hyper-spectral missions.

References

1. S. Murchie, R. Arvidson, P. Bedini, et al., "CRISM (Compact Reconnaissance Imaging Spectrometer for Mars) on MRO," Journal of Geophysical Research: Planets, vol. 112, no. E5, 2007.
2. C. M. Pieters, J. N. Goswami, R. N. Clark, et al., "Character and Spatial Distribution of OH/H_2O on the Surface of the Moon Seen by M^3 on Chandrayaan-1," Science, vol. 326, no. 5952, pp. 568–572, 2009.
3. J. F. Mustard, S. L. Murchie, S. M. Pelkey, et al., "Hydrated Silicate Minerals on Mars Observed by the CRISM Instrument on MRO," Nature, vol. 454, no. 7202, pp. 305–309, 2008.
4. C. E. Viviano-Beck, F. P. Seelos, S. L. Murchie, et al., "Revised CRISM Spectral Parameters and Summary Products Based on the Currently Detected Mineral Diversity on Mars," Journal of Geophysical Research: Planets, vol. 119, no. 6, pp. 1403–1431, 2014.
5. P. Kumari, S. Soor, A. Shetty, and S. G. Koolagudi, "A Fully-Automated Framework for Mineral Identification on Martian Surface Using Supervised Learning Models," IEEE Access, vol. 11, pp. 13121–13137, 2023.
6. R. N. Clark and T. L. Roush, "Reflectance Spectroscopy: Quantitative Analysis Techniques for Remote Sensing Applications," Journal of Geophysical Research: Solid Earth, vol. 89, no. B7, pp. 6329–6340, 1984.
7. R. F. Kokaly, R. N. Clark, G. A. Swayze, et al., "USGS Spectral Library Version 7," U.S. Geological Survey Data Series 1035, 2017.
8. B. Palsson, J. Sigurdsson, J. R. Sveinsson, and M. O. Ulfarsson, "Convolutional Autoencoder for Spectral–Spatial Hyperspectral Unmixing," IEEE Transactions on Geoscience and Remote Sensing, vol. 59, no. 1, pp. 535–549, 2022.

9. T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A Simple Framework for Contrastive Learning of Visual Representations," in Proceedings of the International Conference on Machine Learning (ICML), 2020.
10. L. McInnes, J. Healy, and S. Astels, "HDBSCAN: Hierarchical Density-Based Clustering," Journal of Open Source Software, vol. 2, no. 11, p. 205, 2017.
11. R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC Superpixels Compared to State-of-the-Art Superpixel Methods," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 34, no. 11, pp. 2274–2282, 2012.
12. P. Krähenbühl and V. Koltun, "Efficient Inference in Fully Connected Conditional Random Fields with Gaussian Edge Potentials," Advances in Neural Information Processing Systems (NeurIPS), vol. 24, 2011.